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MODULAR-TYPE NONLINEARITY IN THE MODELING OF TUMOR SPHEROID GROWTH

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A space–time trigger model of tumor growth is considered depending on the concentration of hydrogen ions, oxygen, and tumor cell density at the initial stage of the tumor spheroid development. A system of parabolic equations with a piecewise linear right-hand side of modular type is used to solve this problem. Numerical implementation is carried out in a three-dimensional domain shaped as a cube with an edge of 0.1 mm *on a uniform grid using the method of lines, the Rosenbrock scheme, and the factorization method in the spatial coordinates. The presented model quite well describes the dynamics of variation in the spheroid area at the initial stage of the tumor development depending on time. A distinctive feature of the model is that it reflects both the process of tumor growth into the external environment and the formation of a necrotic core at the tumor center. Based on the presented system of equations, it is possible to design a model that takes both the heterogeneity of the environment and more complex mechanisms of tumorigenesis into account.*

Keywords: proliferation, tumor cell density, hypoxia, spheroid area, necrotic core, system of parabolic equations, modular nonlinearity, small parameter

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1. Introduction

The tumor is characterized by abnormal proliferation (division) of transformed cells that form the tumor tissue due to an imbalance in the mechanisms of the cell growth control. As the tumor core develops, due to high cellular metabolic activity, there is a shift in pH toward an acidic environment, which has an extremely negative effect on untransformed cells. At the same time, the transformed cells, due to dedifferentiation and disturbances in pro- and anti-apoptotic mechanisms, faster adapt to such changes [1]. It is known that the process of metastasis and invasion into adjacent tissues depend on the extracellular pH index and the metabolism of glucose, which play an important role in the variation in the phenotype of tumor cells. In most normal tissues, the extracellular pH index is equal to 7.2–7.4, and for most solid tumors, pH is equal to 6.5–6.9, which stimulates the invasion into adjacent tissues by remodeling them [2]. Various aspects of the mechanisms of diffusion of substances influencing the formation of the tumor tissue, such as oxygen transport and waste disposal, are described in many works including the earliest ones (see, e.g., [3]–[5]).

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As the tumor process develops, the diffusion of nutrients and oxygen into the central core of the tumor decreases thus causing hypoxic phenomena. As hypoxia increases, the aerobic glycolysis becomes anaerobic in transformed cells, which leads to the release of lactic acid into the extracellular environment thus changing the pH value [6] and increasing the tumor growth into the surrounding tissues [7]. At the same time, reaching the critically low level leads to the death of tumor cells and the formation of a necrotic zone inside it. In a living organism, the glycolysis occurring under anaerobic conditions leads to the accumulation of lactate and a decrease in pH. Depending on the oxygen concentration, the state of the cell changes from normoxic to hypoxic, which is under the influence of the following factors: diffusion constants, vascular network density, oxygen utilization rate by the cells, tumor micro-environment, distance from the main vessels, and other factors [8].

One of the important scientific fields is the creation of a model of tumor growth and its invasion and the description of the mechanism linking changes in glucose metabolism with the ability of transformed cells to form invasive neoplasms. One of the first mathematical models of cell invasion based on changes in pH [9] described the space–time development of the tumor and the normal tissue with the concentration of hydrogen ions taken into account. A numerical study allowed the authors to assume the formation of pH gradient at the "tumor-normal tissue" interface and to demonstrate the conditions for the transition from the benign growth to the malignant growth. Subsequently, a number of mathematical models were presented to study various aspects of cell invasion: vascular and avascular tumor dynamics [10], the role of pH and oxygen in the interaction between normal and tumor cell populations [11] and others [1]. The classical mathematical approach to studying the tumor growth process includes the reaction-diffusion partial differential equations [12], [13]. In some works, tumor nodes are considered as multicellular spheroids [14], [15]. So the paper [14] presents a space–time model of multicellular spheroids taking into account the diffusion of nutrients and their consumption by proliferating and resting cells with different chemotactic responses to a nutrient gradient. Numerical solutions show the dependence of the tumor dynamics on the proliferation rate and cell density in the boundary layer.

In this paper, we present a space–time trigger model of tumor growth depending on the concentration of hydrogen ions, oxygen, and the density of tumor cells at the initial stage of the tumor spheroid development. The object for modeling is a spheroid, which, for example, can be grown in a Petri dish [15]. The model includes three fundamental components that provide an adequate description of the main features of tumor growth: tumor cells, oxygen concentration ($[O_2]$), and the pH level (the concentration of $[H^+]$ cations). A sufficient oxygen concentration is necessary for tumor cell proliferation, while with an increase in their number, the cells inside the tumor experience hypoxia, which in turn initiates glycolysis (the oxidation of glucose contained in the cell). As a result of the latter, the pH indicator shifts to the acidic side which ultimately causes the tumor cell death. A distinctive feature of the model is that it reflects both the process of tumor growth into the external environment and the formation of a necrotic zone at the center of the tumor. This is possible due to the use of parabolic-type equations with modular nonlinearity

2. Modular-type nonlinearity

To describe the processes of tumor growth into the external environment and the cell necrosis inside the spheroid, we use a system of parabolic equations with a piecewise linear right-hand side of the so-called "modular type," namely, equations of the form

$$\frac{\partial u}{\partial t} - D\Delta u = \begin{cases} -u + u_{\min}, & u \leqslant u_{\text{thr}}, \\ -u + u_{\max}, & u > u_{\text{thr}}, \end{cases} \tag{1}$$

where u is the sought quantity, D is the diffusion coefficient, and u_{thr} is the threshold value of the desired quantity at which the phase transition occurs.

It was shown in [16], [17] that such equations can have smooth solutions of the form of moving self-similar wave fronts for small values of the diffusion coefficient ($D \ll 1$). In [17], the following expression was obtained for the zero approximation (with respect to a small parameter D) of the autowave front velocity in an isotropic medium:

$$W = \frac{2u_{\text{thr}} - u_{\min} - u_{\max}}{\sqrt{(u_{\text{thr}} - u_{\min})(u_{\max} - u_{\text{thr}})}}. \tag{2}$$

The positive direction of the front motion is from the domain where the values of u are close to $u_{\min}$ towards the domain where the values of u are close to $u_{\max}$. We note that $u = u_{\min}$ is the equilibrium of Eq. (1) in the domain $u \leqslant u_{\text{thr}}$, and $u = u_{\max}$ is the equilibrium in the domain $u > u_{\text{thr}}$. This property allows using Eq. (1) to model the processes of variation in the substance concentration from the minimal possible level to the saturation level.

3. Model of the spheroid development from tumor cells

To describe the processes discussed above, we propose the system of equations

$$\frac{\partial u}{\partial t} - D_u \Delta u = -\begin{cases} T^{-1}u, & [H^+] \geqslant [H^+]_{\text{thr}}, \\ \begin{cases} T^{-1}(u - u_{\max}), & u \geqslant u_{\text{thr}}, \\ -k_1 \cdot u \cdot [O_2] & u < u_{\text{thr}}, \end{cases} & [H^+] < [H^+]_{\text{thr}}; \end{cases} \tag{3}$$

$$\frac{\partial [H^+]}{\partial t} - D_{[H^+]} \Delta [H^+] = -\begin{cases} T^{-1}([H^+] - [H^+]_{\max}), & [H^+] \geqslant [H^+]_{\text{thr}}, \\ \begin{cases} -k_2 \cdot u(t-\tau), & [O_2] < [O_2]_{\text{hyp}}, \\ T^{-1}[H^+] & [O_2] \geqslant [O_2]_{\text{hyp}}, \end{cases} & [H^+] < [H^+]_{\text{thr}}; \end{cases} \tag{4}$$

$$\frac{\partial [O_2]}{\partial t} - D_{O_2} \Delta [O_2] = -\begin{cases} T^{-1}([O_2] - [O_2]_{\min}), & [O_2] \leqslant [O_2]_{\text{thr}}, \\ T^{-1}([O_2] - [O_2]_{\max}) + k_3 \cdot u \cdot [O_2], & [O_2] > [O_2]_{\text{thr}}. \end{cases} \tag{5}$$

Here, u is the tumor cell density normalized by 1, $[H^+]$ is the cation concentration that determines the pH value by the formula $\text{pH} = -\lg [H^+]$ (where $[H^+]$ is measured in mol/L), $[O_2]$ is the oxygen concentration, T is the characteristic time scale, and the factor T^{-1} was introduced to match the dimensions. At the initial time, an agglomeration of approximately $2 \cdot 10^4$ cells is seeded in a Petri dish.

Equation (3) determines the dynamics of the tumor cell density. The first line in the equation models the function identically equal to zero in the domain, where $[H^+] > [H^+]_{\text{thr}}$, i.e., the pH level is unsuitable for the life of cells. This situation is typical of the tumor necrotic core. If the oxygen level is sufficient and the cell density is less than some threshold value u_{thr}, then their active proliferation occurs [15], which is described by the term $k_1 \cdot u \cdot [O_2]$, where k_1 is the rate of cell division. When the threshold density value (equal to 1) is attained, the cells stop to divide and their maximal density $u_{\max} = 1$ is attained in the domain $u \geqslant u_{\text{thr}}$. To model this saturation state, the term $T^{-1}(u - u_{\max})$ is introdiced in the second line in the right-hand side of Eq. (3).

Equation (4) describes the pH dynamics. In the cells seeded at the initial time, anaerobic glycolysis is activated not immediately but after a short time, which is taken into account in the equation as the argument with a time delay in the term $k_2 \cdot u(t-\tau)$ responsible for the glycolysis. Here, k_2 is the glycolysis coefficient, the dimension of this quantity is T^{-1}. The delay time is the time after which glycolysis begins in the seeded cells, i.e., it is inversely proportional to the glycolysis constant: $\tau = 1/k_2$. The anaerobic glycolysis occurs under hypoxia conditions [18], i.e., when the oxygen concentration is less than the threshold value $[O_2]_{\text{hyp}}$. If the oxygen concentration exceeds the threshold value, then the glycolysis is not anaerobic and the concentration of $[H^+]$ ions remains minimal ($\approx 10^{-7}$). This is described by the third line in the right-hand side of Eq. (4), which models the zero solution. The first line in the right-hand side of Eq. (4) models the attainment of the maximal limit concentration of $[H^+]$ in the necrotic zone (pH ≈ 3).

Equation (5) describes the dynamics of the oxygen concentration. In the second line of the right-hand side of the equation, the combination $-T^{-1}([O_2]-[O_2]_{\max})-k_3 \cdot u \cdot [O_2]$ describes the oxygen consumption by proliferating cells in the case where the oxygen concentration is sufficiently large, $[O_2] > [O_2]_{\text{thr}}$. The first line in the right-hand side of Eq. (5) models the attainment of the minimal possible concentration $[O_2]_{\min}$ in the domain, where the cell density is maximal.

4. Calibration of the model. Choice of parameters

To determine characteristic scales of space and time, we used the data in [15], where an experiment on growing neuroblastoma cells in a Petri dish was carried out. For the numerical experiment, by model (3)–(5), a computational domain shaped as a cube with edge of $R = 0.1$ mm was used.

The initial distribution of cells was taken as a homogeneous spheroid of diameter 0.014 mm with the maximum possible density of cells equal to 1. The initial distribution of the oxygen concentration was taken equal to $[O_2]_{\max} = 0.0556$ mmol/L [15] in the domain, where there are no cells, and equal to $[O_2]_{\min} = 0.05 \cdot [O_2]_{\max}$ at the site of the initial agglomeration of cells [19]. The initial ion concentration $[H^+]$ was taken equal to zero, which practically corresponds to the level pH = 7.2–7.4. On all boundaries of the computational domain, the homogeneous Neumann condition was posed.

The threshold pH level below which the necrosis of tumor cells begins was taken equal to 3.022, and the minimal possible pH level was equal to 3, which corresponds to the value of ion concentration $[H^+]$ in the saturated aqueous solution (1 mmol/L). The threshold level of the oxygen concentration determining the hypoxia state is $[O_2]_{\text{hyp}} = 0.2 \cdot [O_2]_{\max}$ [20]. The threshold level of the oxygen concentration at which the active proliferation of cells begins is $[O_2]_{\text{thr}} = 0.036$ mmol/L. This value was taken using formula (2) for the front velocity. By this formula, to model the oxygen concentration distribution in the form of a step with the minimum value at the tumor center, the maximum value far from the cell localization site, and a large gradient at the edge of the expanding tumor, the threshold value of oxygen concentration must be greater than half the sum of the minimum and maximum values of this concentration. The threshold value u_{thr} was chosen equal to 0.05, i.e., 5% of the maximum value of cell concentration.

According to the experimental data from [15], the initial linear size of the tumor was 0.01 mm, and after seven days the surface area of the spheroid was more than 0.01 mm^2. Comparing the tumor growth time from the experiment with the dimensionless time used in the model under similar initial conditions, we see that the dimensionless time unit T of the model is equal to 19 hours.

As was noted in [17], the solutions of the moving front type in equations with modular nonlinearity arise only for sufficiently small factors at the second derivative with respect to the coordinate. As a result of numerical experiments, the following coefficients were chosen: $D_{O_2} = 0.0002\ R^2/T \approx 3 \cdot 10^{-11}$ mm^2/s; $D_H = 0.0001\ R^2/T \approx 1.5 \cdot 10^{-11}$ mm^2/s; $D_u = 0.00001\ R^2/T \approx 1.5 \cdot 10^{-12}$ mm^2/s. These coefficients have the meaning of quantitative measure of the tumor ability to grow under given conditions, i.e., they reflect the influence of the environment on the tumor cells.

The coefficient k_1 corresponds to the cell division rate and was chosen based on the data on the time of doubling the population, which varies from 12 to 24 hours [15]. The time unit of the model is 19 hours. In numerical calculations, the value $k_1 = 1$ was used which corresponds to the rate of cell division once in 19 hours.

The physical meaning of the coefficient k_2 is the rate of glycolysis. The aerobic glycolysis in a cell occurs during 2–3 hours [21]. If the unit of time in the model is 19 hours, then the coefficient k_2 must be 6–8 T^{-1}. In the numerical implementation, it was taken as $k_2 = 6$.

The coefficient k_3 reflects the oxygen homeostasis of the cell. In physical terms, it should be proportional to the rate of oxygen absorption by the cell during the cell division. In the model, this coefficient was chosen based on the ratio of a sufficient amount of oxygen in the organelle, corresponding to the partial pressure pO_2 from 20 to 60 mm Hg and higher (the oxygen level in the culture medium) to a partial pressure pO_2 in the range from 5 to 20 mm Hg in the state of hypoxia in the organelle [19]. In numerical calculations, it was taken as $k_3 = 4$, which agrees with the physically permissible range of the specified ratio.

5. Results of the numerical experiment

The numerical implementation was carried out in a three-dimensional domain shaped as a cube with side $R = 0.1$ mm on a uniform grid with step $R/140$ in each direction using the method of straight lines with time step 0.002, the Rosenbrock scheme with the coefficient 1/2, and the factorization by spatial coordinates [22]. When numerically solving system (3)–(5), we used the parameters of the space–time grid justified in the previous investigation [23], in which a scalar equation with modular nonlinearity was considered. Distributed computing on the CUDA compiler was used in the calculations.

Figure 1 shows the dynamics of the distribution of the normalized cell density (left column), oxygen concentration (middle column), and cation concentration (right column) in the central section of the spheroid: a, b are initial distribution (there are no $[H^+]$ ions at the initial time); c, d, e are those 24 hours after seeding the culture; f, g, h, after three days; and j, k, l, after seven days. We can see the formation of various zones of the spheroid growth. On one hand, the appearance of a necrotic core can be observed. On the other hand, we can see transition layers of tumor cells with different proliferation rates due to changes in oxygen concentration and the pH indicator in the boundary zones. At the initial time, when the cell colony is small, the oxygen concentration is sufficient to maintain the vital activity of transformed cells in the entire volume of the spheroid. Subsequently, as the spheroid grows, the oxygen concentration in its central part falls below the threshold value, which leads to the death of tumor cells and the formation of a necrotic core.

The presented model quite well describes the dynamics of the spheroid growth, which is consistent with known experimental data. Figure 2a illustrates the comparison of graphs of the increase in the surface area of the spheroid (in mm^2) depending on time (in days). The asterisks mark the results of the numerical experiment. The curve with experimental points is a graph from [15].

Figure 2b shows a three-dimensional illustration of the spheroid in a cross section through the central plane after seven days after seeding the culture: the dark color indicates the zone in which the cell density is at least 0.8, and the light color show the zone, where it is less than 0.8 (transition layers).

6. Discussion

• The model quite well describes the dynamics of changes in the area of the spheroid depending on the time of development of the tumor process (Fig. 2a).

• The use of modular nonlinearity made it possible to create a model of tumor cell development reflecting both the process of tumor growth into the external environment and the formation of a necrotic zone in the center of the tumor. An attempt to create a model describing both of these effects was made in [24]. However, the authors of that work note that the effect of the existence of a necrotic zone is present only at the initial time and is not preserved in a sufficiently long period of numerical calculation.

• The presence of a small parameter at the highest derivative in Eqs. (3)–(5) allows an asymptotic analysis of the presented system of equations in order to clarify the applicability limits of the models and the front motion velocities; however, the corresponding study is the topic of a separate publication planned by the authors in the future. In the framework of this study, we used the already obtained result of asymptotic analysis given in [17] for the scalar case, in particular, this result was used to obtain formula (2) for the velocity.

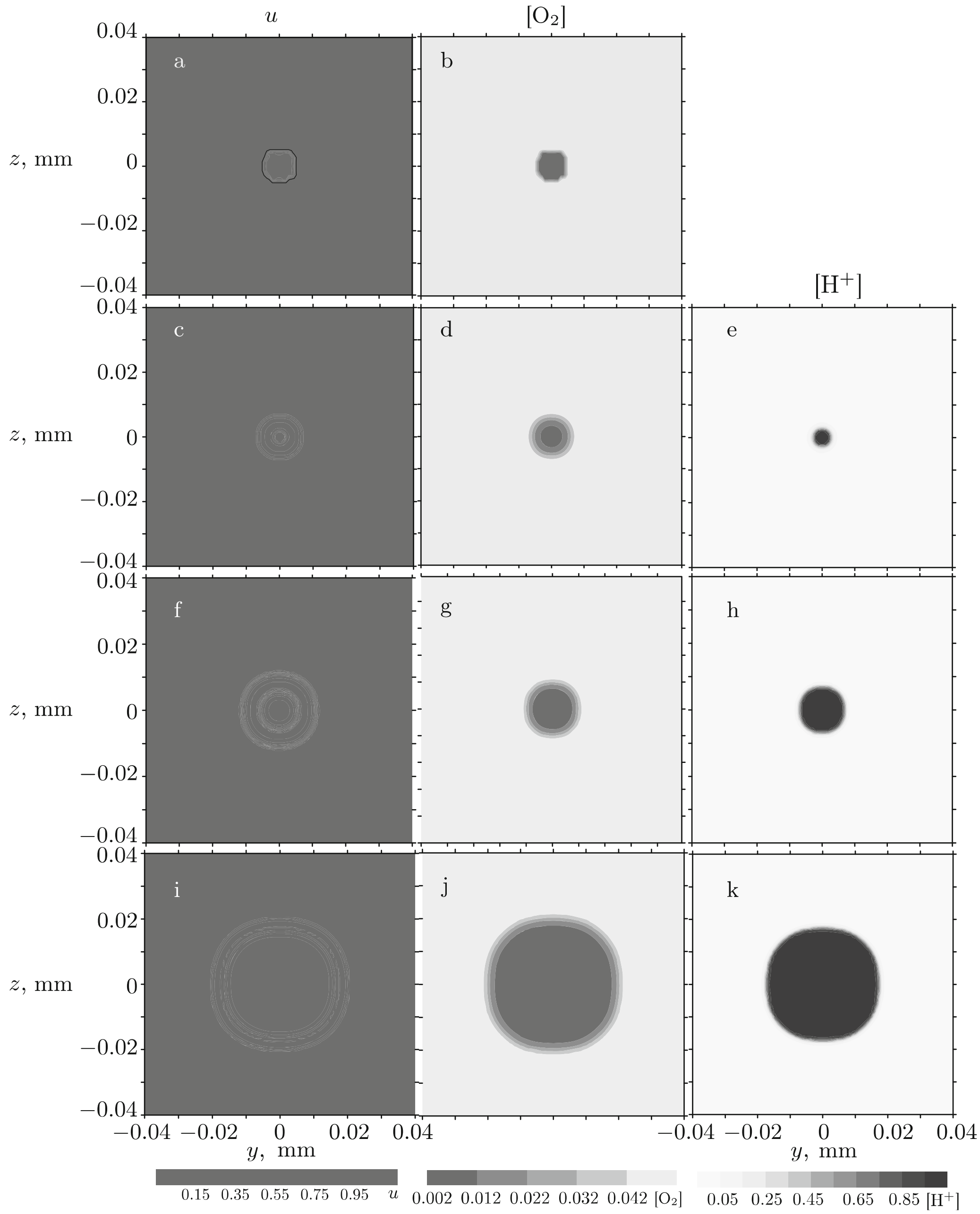

Fig. 1. Dynamics of the distribution of the normalized cell density (left column), oxygen concentration (middle column), and cation concentration (right column) in the central section of the spheroid: (a, b) the initial distribution (there are no $[H^+]$ ions at the initial time); (c, d, e) 24 hours after seeding the culture; (f, g, h) after three days; and (j, k, l) after seven days.

• The parameters k_1, k_2, k_3, combined with the threshold values $[O_2]_{\rm thr}$, $[H^+]_{\rm thr}$, and $u_{\rm thr}$, determine the values of the velocities of the fronts of tumor spread into the external environment and the expansion of the necrotic zone inside the tumor. Different values of the parameters correspond to different phenotypes of tumor cells.

• For parabolic equations with modular nonlinearity, the existence of solutions in the form of moving fronts is possible only for sufficiently small parameters D_u, D_H, and D_{O_2} [17]. We note that in models

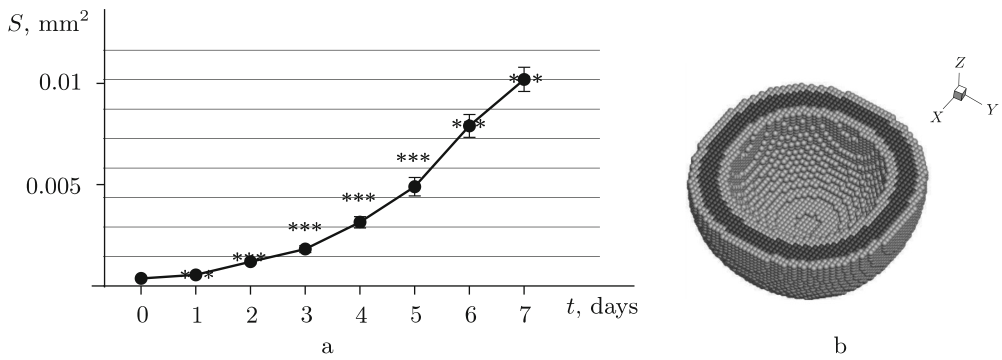

Fig. 2. (a) Result of the comparison of the spheroid area (mm^2) dependence on time (days). The asterisks mark the results of the numerical modeling, the curve with marked experimental points is a graph from [15]. (b) Three-dimensional illustration of the spheroid in a cross section through the central plane after seven days after seeding the culture: the dark color indicates the zone in which the cell density is at least 0.8, and the light color show the zone, where it is less than 0.8 (transition layers).

often found in the literature [15], [25], diffusion equations with coefficients of oxygen and cations diffusion of the order of 10^{-2} mm^2/s are used as a rule. In [26], the experimentally obtained coefficient of oxygen diffusion in multicellular spheroids is given, which is equal to 10^{-3} mm^2/s. Such a diffusion coefficient was used in the models studied in [8], [20]. Some models also use equations with a nonlinear diffusion coefficient depending on the density of tumor cells [12], [27]. However, the model presented in this paper can be implemented only for significantly smaller values of the diffusion coefficients. In connection with this, we calibrated the model based on the experimental value of the growth rate of the spheroid from [15]. In the proposed model, the coefficients D_u, D_H, and D_{O_2} reflect the influence of the environment on the tumor cells. At large values of these coefficients, system (3)–(5) does not have a solution of the front type, i.e., the tumor is not formed. At sufficiently small values of the coefficients D_u, D_H, and D_{O_2}, the formation of agglomerations of tumor cells is possible. Thus, the variation of these coefficients can be used in modeling the action of the therapy.

• To improve the model of tumor node development in a living organism in the future, it is necessary to account for the influence of the process of angiogenesis, remodulation of the intercellular matrix, infiltration of the tumor by the lymphoid tissue, the presence or absence of hormonal receptors, specific therapy, and many other factors. The coefficients and dimensions should be selected taking these mechanisms of tumorigenesis into account. Mathematical modeling with more complex mechanisms will allow testing potential antitumor drugs for their effectiveness.

7. Conclusions

This work is the initial stage of creating a model of cancer development pursued by the team of authors of the paper. At the first stage, the main components of the model are identified, which allows describing the main characteristics of tumor development *in vitro*, comparing with the results of the experiment in [15], and calibrating the time and space scales. Based on system (3)–(5), it is possible to create a model that takes the heterogeneity of the environment and initial conditions into account and also directly accounts for the development mechanisms of the tumor itself, which leads to the formation of the tumor boundary and the angiogenesis in a more distant perspective.

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Conflict of interest. The authors of this work declare that they have no conflicts of interest.

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